Hyperspectral and Multispectral Image Fusion via Variational Tensor Subspace Decomposition

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***Abstract*— The fusion of hyperspectral image (HSI) and multispectral image (MSI) refers to enhance the spatial resolution of HSI with the help of a corresponding MSI that has a high spatial resolution to finally obtain an HSI with high resolution in both spatial and spectral domains. In this letter, we propose a variational tensor subspace decomposition-based fusion method to fully explore the differences and correlations among three modes of the HSI tensor. Experimental results on two HSI datasets show that the proposed method can achieve superior performance compared with existing state-of-the-art fusion methods with high computational efficiency.**

***Index Terms*— Hyperspectral, image fusion, tensor decomposition, total-variation (TV).**

I. Introduction

HYPERSPECTRAL images (HSIs) are usually composed of hundreds of narrow bands, which make them have a high spectral resolution. However, due to the limitations of imaging systems, they always have a low spatial resolution. Contrarily, multispectral images (MSIs) have high spatial resolution and relatively lower spectral resolution compared with HSIs. Many applications [1], [2] require high spectral and high spatial resolution data, which motivate researchers to look for some algorithmic approaches to fuse HSI and MSI.

Early research studies generalized the fusion algorithms of panchromatic and MSI, such as principal component analysis [3] and wavelet transform [4], to realize the fusion of MSI and HSI, but their resample operations on the spectral bands lead to spectral distortions of the results [5]. Concerning that the observed data, namely the low-resolution (LR) HSI and the high-resolution (HR) MSI, are degraded versions of the unknown HR-HSI, there are many reconstruction-based fusion algorithms. Because the reconstruction is an ill-posed problem, some priors of HR-HSIs are always explored [6], [7]. Among them, total-variation (TV) prior is widely used. Simões *et al*. [7] used the vector-based-TV to obtain the spatial smoothness. But it could not obtain a satisfactory result because the spatial and spectral data are first vectorized, which destroyed the intrinsic structure of images. Wang *et al*. [8] constructed a nonconvex tensor penalty to exploit the intrinsic characteristics in 4-D tensor, and explored the local smoothness in both spatial and spectral modes by 3-D TV term. But it first clustered tensors to some groups, which brought extra computation complexity. Xu *et al*. [9] proposed a unidirectional TV-based method, and considered the target HR-HSI exhibited the piecewise smoothness on the three modes. However, the TV regularizations were only imposed in three dictionaries.

Very recently, some deep learning (DL)-based fusion methods were proposed. Dian *et al*. [10] proposed a method that utilized traditional optimization as the basic framework and a convolutional neural network (CNN) to learn the priors. Palsson *et al*. [11] used a 3-D CNN to fuse HSI and MSI. Zhang *et al*. [12] proposed a spatial-spectral reconstruction network (SSR-Net) for the fusion of HSI and MSI. Although achieving good performances, DL-based methods usually need massive amounts of training data, which is often unpractical in the fusion problem.

There are two problems in the TV-based fusion methods. First of all, most of them explored the spatial and spectral smoothness in the original image space, which leads to high computational costs. Based on the assumption that spectral vectors live in a low-dimensional subspace, the low tensor multirank (LTMR) method [13] learned the spectral subspace from the LR-HSI, and estimated the coefficients via LTMR prior. The second one is that available methods only utilized the subspace learning in the spectral domain to reduce the computational costs, regardless of the spatial dimension. Actually, searching for a low-dimensional subspace in the spatial domain is difficult. Although 3-D TV regularization is always explored to model the spatial-spectral smoothness [14], it neglects the correlations among them. We compute the gradient tensors of three modes, and then calculate the correlation coefficients (CCs) among them, the distribution of the CCs is shown in the first row of Fig. 1. From these figures, we can observe that the gradient tensors are correlated with each other, especially between the horizontal and vertical gradients. Therefore, we should project them to space where the common and the different parts can be separated as soon as possible, meanwhile, this space should be low-dimensional.

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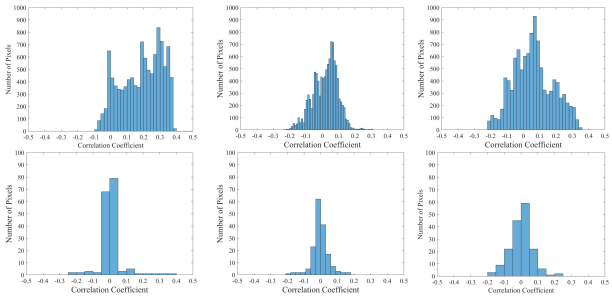

Fig. 1. Distributions of CCs, where the first row shows the CCs calculated in original gradient space, and the second row shows CCs calculated in the projected subspace. From left to right, CCs between horizontal and vertical, horizontal and spectral, and vertical and spectral gradients.

Aiming at solving the problems mentioned above, in this letter, the subspace learning is incorporated into the Tucker decomposition framework to formulate a variational tensor subspace decomposition method for the reconstruction of HR-HSIs. As the second row of Fig. 1 shows, the correlations of three gradient tensors are fully reduced. Hence, the sparsity regularization on the gradient maps can be conducted in the projected subspace.

II. Proposed Method

In this section, we first review some basics in tensor multilinear algebra, then formulate our method. Finally, we develop an efficient algorithm based on the alternating direction method of multipliers (ADMMs) to solve the problem.

A. Tensor Basics

We use the Euler script letters to represent the tensors. An Nth-order tensor is denoted as $\mathcal{A} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ and its elements are $a_{i_1,i_2,\ldots,i_N}$, where $1 \leq i_n \leq I_n$, $n = 1, 2, \ldots, N$. The Frobenius norm of $\mathcal{A}$ is defined by $\|\mathcal{A}\|_F = \left(\sum_{i_1,i_2,\ldots,i_N} a^2_{i_1,i_2,\ldots,i_N}\right)^{1/2}$, and the ℓ_1-norm of $\mathcal{A}$ is $\|\mathcal{A}\|_1 = \sum_{i_1,i_2,\ldots,i_N} |a_{i_1,i_2,\ldots,i_N}|$. The mode-$n$ matricization, also known as the unfolding of the tensor, is represented as $\mathbf{A}_{(n)} =: \text{unfold}_n(\mathcal{A}) \in \mathbb{R}^{I_n \times (I_1,\ldots,I_{n-1} I_{n+1},\ldots,I_N)}$, and it can also be transformed to original tensor format by $\mathcal{A} =: \text{fold}_n(\mathbf{A}_{(n)})$. Then the n-mode product of $\mathcal{A}$ with a matrix $\mathbf{U} \in \mathbb{R}^{J \times I_n}$ is $\mathcal{B} = \mathcal{A} \times_n \mathbf{U} \in {}^{I_1 \times \cdots \times I_{n-1} \times J \times I_{n+1} \times \cdots \times I_N}$, and it can also be calculated by the matrix multiplication $\mathbf{B}_{(n)} = \mathbf{U}\mathbf{A}_{(n)}$, therefore, the tensor $\mathcal{B}$ can be obtained by $\mathcal{B} = \text{fold}_n(\mathbf{B}_{(n)})$ [15].

Definition 1 (Tucker Decomposition): A Tucker decomposition of tensor $\mathcal{A} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ is the decomposition of $\mathcal{A}$ as a core tensor multiplied by a matrix along each mode

$$\mathcal{A} = \mathcal{C} \times_1 \mathbf{L}_1 \times_2 \mathbf{L}_2 \cdots \times_N \mathbf{L}_N \tag{1}$$

where $\mathbf{L}_n \in \mathbb{R}^{I_n \times J_n}$ are the factor matrices along each mode, and $\mathcal{C} \in \mathbb{R}^{J_1 \times J_2 \times \cdots \times J_N}$ is the core tensor. The mode-n unfolding of the Tucker decomposition is

$$\mathbf{A}_{(n)} = \mathbf{L}_n \mathbf{C}_{(n)} (\mathbf{L}_N \otimes \cdots \otimes \mathbf{L}_{n+1} \otimes \mathbf{L}_{n-1} \otimes \cdots \otimes \mathbf{L}_1)^T \tag{2}$$

where "$\otimes$" represents the Kronecker product.

B. Problem Formulation

The fusion problem aims to obtain the HR-HSI from LR-HSI and HR-MSI. Let $\mathcal{X} \in \mathbb{R}^{H \times W \times s}$, $\mathcal{Y} \in \mathbb{R}^{h \times w \times S}$, and $\mathcal{Z} \in \mathbb{R}^{H \times W \times S}$ represent the HR-MSI, LR-HSI and HR-HSI, respectively, where $H = h \times r$, $W = w \times r$ (r is the ratio between their spatial resolution), and s (S) are the number of spectral bands of MSI (HSI). First of all, HR-MSI is considered as a degraded version of HR-HSI in the spectral domain [7], [13], then we have

$$\mathcal{X} = \mathcal{Z} \times_3 \mathbf{R} \tag{3}$$

where $\mathbf{R} \in \mathbb{R}^{s \times S}$ is the spectral response matrix. Similarly, LR-HSI is the spatially degraded version of HR-HSI [7], [13]

$$\mathcal{Y} = \mathcal{Z} \times_1 \mathbf{P}_1 \times_2 \mathbf{P}_2 \tag{4}$$

where $\mathbf{P}_1 \in \mathbb{R}^{h \times H}$ and $\mathbf{P}_2 \in \mathbb{R}^{w \times W}$ are the spatial downsampling and blurring matrices along the height and width modes.

C. Variational Tensor Subspace Decomposition

There are various TV types in the field of HSIs processing. Among them, the anisotropic spatial-spectral TV (SSTV) regularization is widely used in recent years, and it is defined as follows: [14]:

$$\|\mathcal{Z}\|_{\text{SSTV}} = \sum_n \|\nabla_n \mathcal{Z}\|_1 \tag{5}$$

where $\nabla_n \mathcal{Z}$ denotes the first-order forward finite-difference operators in $\mathcal{Z}$ along the mode-n, i.e., gradient map of mode-n. Though the SSTV regularization considers the local smoothness both in the spatial and spectral domains, it constrains the gradient maps regardless of their correlations and differences, which motivates us to explore the intrinsic characteristics of HSIs.

Based on the Tucker decomposition given in (1), we have the following proposition.

Proposition 1: If a tensor $\mathcal{A}$ can be decomposed by the Tucker decomposition shown in (1), the following equation will always hold:

$$\text{unfold}_n(\nabla_n \mathcal{A}) = \mathbf{D}_n * \mathbf{A}_{(n)}, \quad n = 1, \ldots, N \tag{6}$$

where $\mathbf{D}_n$ is the difference operator along mode-n, and "$*$" represents the convolution operation.

According to the definition of Tucker decomposition, the HR-HSI $\mathcal{Z}$ can be decomposed as

$$\mathcal{Z} = \mathcal{S} \times_1 \mathbf{U}_1 \times_2 \mathbf{U}_2 \times_3 \mathbf{U}_3 \tag{7}$$

where $\mathcal{S} \in \mathbb{R}^{I \times J \times K}$ is the core tensor. $\mathbf{U}_1 \in \mathbb{R}^{H \times I}$, $\mathbf{U}_2 \in \mathbb{R}^{W \times J}$ and $\mathbf{U}_3 \in \mathbb{R}^{S \times K}$ are the factor matrices along three modes. We have

$$\begin{aligned} \mathbf{Z}_{(1)} &= \mathbf{U}_1 \mathbf{S}_{(1)} (\mathbf{U}_3 \otimes \mathbf{U}_2)^T \\ \mathbf{Z}_{(2)} &= \mathbf{U}_2 \mathbf{S}_{(2)} (\mathbf{U}_3 \otimes \mathbf{U}_1) \\ \mathbf{Z}_{(3)} &= \mathbf{U}_3 \mathbf{S}_{(3)} (\mathbf{U}_2 \otimes \mathbf{U}_1). \end{aligned} \tag{8}$$

Let $\mathbf{M}_1 = \mathbf{D}_1 \mathbf{U}_1 \mathbf{S}_{(1)}$, $\mathbf{M}_2 = \mathbf{D}_2 \mathbf{U}_2 \mathbf{S}_{(2)}$, $\mathbf{M}_3 = \mathbf{D}_3 \mathbf{U}_3 \mathbf{S}_{(3)}$, $\mathbf{V}_1 = \mathbf{U}_3 \otimes \mathbf{U}_2$, $\mathbf{V}_2 = \mathbf{U}_3 \otimes \mathbf{U}_1$, and $\mathbf{V}_3 = \mathbf{U}_2 \otimes \mathbf{U}_1$, then we have

$$\begin{aligned} \|\mathcal{Z}\|_{\text{SSTV}} &= \sum_n \|\nabla_n \mathcal{Z}\|_1 = \sum_n \|\text{unfold}_n(\nabla_n \mathcal{Z})\|_1 \\ &= \sum_n \left\|\mathbf{D}_n * \mathbf{Z}_{(n)}\right\|_1 = \sum_n \left\|\mathbf{M}_n \mathbf{V}_n^T\right\|_1. \end{aligned} \tag{9}$$

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If we constrain $\mathbf{V}_n$ as orthogonal matrices, the projected subspaces of three modes are separated to each other. Therefore, combined with (3) and (4), our reconstruction problem is converted to a constrained minimization problem

$$\begin{aligned}\min_{\mathcal{Z},\mathbf{M}_n,\mathbf{V}_n} & \frac{1}{2}\|\mathcal{X}-\mathcal{Z}\times_3\mathbf{R}\|_F^2+\frac{1}{2}\|\mathcal{Y}-\mathcal{Z}\times_1\mathbf{P}_1\times_2\mathbf{P}_2\|_F^2\\ & +\lambda\sum_{n=1}^{3}\|\mathbf{M}_n\|_1\\ \text{s.t. } & \mathbf{D}_n*\mathbf{Z}_{(n)}=\mathbf{M}_n\mathbf{V}_n^T,\quad \mathbf{V}_n^T\mathbf{V}_n=\mathbf{I}.\end{aligned}\tag{10}$$

To solve the problem based on ADMM, we first introduce an auxiliary variable $\mathcal{H}=\mathcal{Z}$, and then utilize the augmented Lagrangian method to obtain the following objective function:

$$\begin{aligned}&\mathcal{L}(\mathcal{Z},\mathcal{H},\mathbf{M}_n,\mathbf{V}_n,\mathbf{T}_n,\mathcal{J}_1)\\ &=\frac{1}{2}\|\mathcal{X}-\mathcal{Z}\times_3\mathbf{R}\|_F^2\\ &\quad+\frac{1}{2}\|\mathcal{Y}-\mathcal{Z}\times_1\mathbf{P}_1\times_2\mathbf{P}_2\|_F^2+\lambda\sum_{n=1}^{3}\|\mathbf{M}_n\|_1\\ &\quad+\sum_{n=1}^{3}\left\langle\mathbf{T}_n,\mathbf{D}_n\mathbf{H}_{(n)}-\mathbf{M}_n\mathbf{V}_n^T\right\rangle+\langle\mathcal{H}-\mathcal{Z},\mathcal{J}_1\rangle\\ &\quad+\frac{\alpha}{2}\sum_{n=1}^{3}\left\|\mathbf{D}_n\mathbf{H}_{(n)}-\mathbf{M}_n\mathbf{V}_n^T\right\|_F^2+\frac{\beta}{2}\|\mathcal{H}-\mathcal{Z}\|_F^2\end{aligned}\tag{11}$$

where $\mathbf{T}_n$, $(n=1,2,3)$ and $\mathcal{J}_1$ are the Lagrangian multipliers and α and β are the positive scalars. Now, we can solve the problem by alternatively optimizing each variable with other variables fixed.

1) Optimizing $\mathcal{Z}$: The optimization function with respect to $\mathcal{Z}$ is

$$\begin{aligned}\widehat{\mathcal{Z}}=\arg\min_{\mathcal{Z}}&\frac{1}{2}\|\mathcal{X}-\mathcal{Z}\times_3\mathbf{R}\|_F^2+\frac{1}{2}\|\mathcal{Y}-\mathcal{Z}\times_1\mathbf{P}_1\times_2\mathbf{P}_2\|_F^2\\ &+\langle\mathcal{H}-\mathcal{Z},\mathcal{J}_1\rangle+\frac{\beta}{2}\|\mathcal{H}-\mathcal{Z}\|_F^2.\end{aligned}\tag{12}$$

According to the Tucker decomposition, (12) can be reformulated as

$$\begin{aligned}\widehat{\mathcal{Z}}=\arg\min_{\mathcal{Z}}&\frac{1}{2}\left\|\mathbf{X}_{(3)}-\mathbf{R}\mathbf{Z}_{(3)}\right\|_F^2+\frac{1}{2}\left\|\mathbf{Y}_{(3)}-\mathbf{Z}_{(3)}\mathbf{P}\right\|_F^2\\ &+\frac{\beta}{2}\left\|\mathcal{H}-\mathcal{Z}+\frac{\mathcal{J}_1}{\beta}\right\|_F^2\end{aligned}\tag{13}$$

where $\mathbf{P}=\mathbf{P}_1\otimes\mathbf{P}_2\in\mathbb{F}^{HW\times hw}$. Solving (13) is equivalent to solving the following linear system:

$$\mathbf{B}_1\mathbf{Z}_{(3)}+\mathbf{Z}_{(3)}\mathbf{B}_2=\mathbf{B}_3\tag{14}$$

where $\mathbf{B}_1=\mathbf{R}^T\mathbf{R}+\beta\mathbf{I}$, $\mathbf{B}_2=\mathbf{P}\mathbf{P}^T$, and $\mathbf{B}_3=\mathbf{Y}_{(3)}\mathbf{P}^T+\mathbf{R}^T\mathbf{X}_{(3)}+(\beta\mathbf{H}_{(3)}+\mathbf{J}_{1(3)})$. According to Wei *et al.* [6], (14) is a Sylvester matrix equation, and has a unique solution. Therefore, we can obtain $\mathcal{Z}$ through the algorithm given in [6].

2) Optimizing $\mathcal{H}$: Keeping other variables fixed, the optimization function with respect to $\mathcal{H}$ is

$$\begin{aligned}\widehat{\mathcal{H}}=\arg\min_{\mathcal{H}}&\sum_{n=1}^{3}\left\langle\mathbf{T}_n,\mathbf{D}_n\mathbf{H}_{(n)}-\mathbf{M}_n\mathbf{V}_n^T\right\rangle\\ &+\frac{\alpha}{2}\sum_{n=1}^{3}\left\|\mathbf{D}_n\mathbf{H}_{(n)}-\mathbf{M}_n\mathbf{V}_n^T\right\|_F^2+\frac{\beta}{2}\left\|\mathcal{H}-\mathcal{Z}+\frac{\mathcal{J}_1}{\beta}\right\|_F^2.\end{aligned}\tag{15}$$

Deriving (15) with respect to $\mathcal{H}$ in the Fourier domain, we can obtain its solution

$$\mathcal{H}=\mathbb{F}^{-1}\left(\frac{\mathbb{F}(\beta\mathcal{Z}-\mathcal{J}_1)+\sum_{n=1}^{3}\mathbb{F}^*(\mathbf{D}_n)\left(\mathbb{F}\left(\alpha\mathbf{M}_n\mathbf{V}_n^T-\mathbf{T}_n\right)\right)}{\beta+\alpha\sum_{n=1}^{3}\mathbb{F}^*(\mathbf{D}_n)\mathbb{F}(\mathbf{D}_n)}\right)\tag{16}$$

where $\mathbb{F}(\cdot)$ and $\mathbb{F}^{-1}(\cdot)$ are the Fourier transform and the inverse Fourier transform, and $\mathbb{F}^*(\mathbf{D}_n)$ represents the conjugate of $\mathbb{F}(\mathbf{D}_n)$.

3) Optimizing $\mathbf{M}_n$: Fixing other variables, we have the following optimization function with respect to $\mathbf{M}_n$:

$$\widehat{\mathbf{M}}_n=\arg\min_{\mathbf{M}_n}\lambda\|\mathbf{M}_n\|_1+\frac{\alpha}{2}\left\|\mathbf{M}_n-\left(\mathbf{D}_n\mathbf{H}_{(n)}+\frac{\mathbf{T}_n}{\alpha}\right)\mathbf{V}_n\right\|_F^2.\tag{17}$$

The subproblem can be solved by employing the soft-thresholding algorithm [16]

$$\mathbf{M}_n=\mathbb{S}_{\frac{\lambda}{\alpha}}\left(\left(\mathbf{D}_n\mathbf{H}_{(n)}+\frac{\mathbf{M}_n}{\alpha}\right)\mathbf{V}_n\right)\tag{18}$$

where $\mathbb{S}$ represents the soft-thresholding operator.

4) Optimizing $\mathbf{V}_n$: We achieve the following optimization function for the variable $\mathbf{V}_n$:

$$\begin{aligned}\widehat{\mathbf{V}}_n&=\arg\min_{\mathbf{V}_n}\frac{\alpha}{2}\left\|\mathbf{M}_n\mathbf{V}_n^T-\left(\mathbf{D}_n\mathbf{H}_{(n)}+\frac{\mathbf{T}_n}{\alpha}\right)\right\|_F^2\\ &=\arg\min_{\mathbf{V}_n}\left\langle\left(\mathbf{D}_n\mathbf{H}_{(n)}+\frac{\mathbf{T}_n}{\alpha}\right)^T\mathbf{M}_n,\mathbf{V}_n\right\rangle.\end{aligned}\tag{19}$$

Let $\mathbf{F}_n=(\mathbf{D}_n\mathbf{H}_{(n)}+\frac{\mathbf{T}_n}{\alpha})^T\mathbf{M}_n$, and we have its condensed singular value decomposition (SVD) $\mathbf{F}_n=\mathbf{G}_n\mathbf{Q}_n\mathbf{W}_n^T$. According to the theorem provided in [17], if we have $\mathbf{V}_n^T\mathbf{V}_n=\mathbf{I}$, the global solution of (19) is

$$\mathbf{V}_n=\mathbf{G}_n\mathbf{W}_n^T.\tag{20}$$

Then we obtain the optimized $\mathbf{V}_n$.

Finally, the Lagrangian multipliers are updated by

$$\begin{aligned}\mathbf{T}_n&:=\mathbf{T}_n+\alpha\left(\mathbf{D}_n\mathbf{H}_{(n)}-\mathbf{M}_n\mathbf{V}_n^T\right)\\ \mathcal{J}_1&:=\mathcal{J}_1+\beta(\mathcal{H}-\mathcal{Z}).\end{aligned}\tag{21}$$

The overall optimization process is shown in Algorithm 1.

III. Experiments

In this section, we compare the proposed method with several related fusion methods including hyperspectral image super-resolution method (HySure) [7], super-resolution tensor-reconstruction (STEREO) method [18], LTMR method [13], and SSR-Net [12], all the parameters are chosen according to the authors' advice. Due to the page limitation, we conduct experiments on a synthetic dataset and a real dataset. In order to make a numerical comparison, the peak signal-to-noise ratio (PSNR), the relative dimensionless global error in synthesis (ERGAS), the average spectral angle mapper (SAM), and the universal image quality index (UIQI) [19] are used as the quality measures in the synthetic experiment. Moreover, the running time of all methods is also shown in Table I. All the methods except for SSR-Net [12] are

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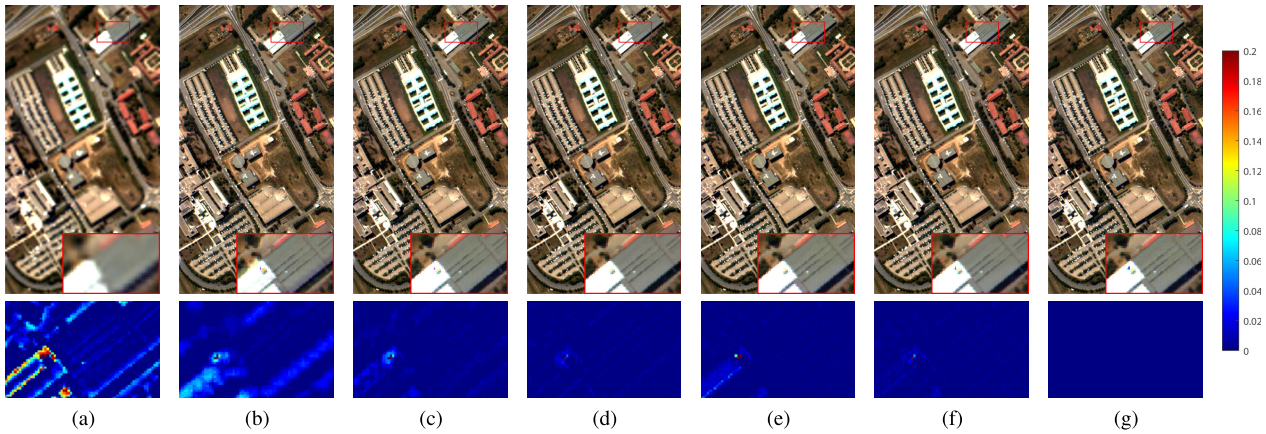

Fig. 2. Experimental results and difference maps of Pavia University data. (a) LR-HSI. (b) HySure [7]. (c) STEREO [18]. (d) LTMR [13]. (e) SSR-Net [12]. (f) Proposed. (g) Reference.

Algorithm 1 Proposed Method

Input: HR-MSI $\mathcal{X}$, LR-HSI $\mathcal{Y}$, spectral response matrix $\mathbf{R}$, and the spatial degradation matrix $\mathbf{P}$

Initialize: $\mathcal{J}_1^{(0)}$, $\mathcal{Z}^{(0)}$, $\mathcal{H}^{(0)}$, $\mathbf{M}_n^{(0)}$, $\mathbf{V}_n^{(0)}$, $\mathbf{T}_n^{(0)}$, $\alpha^{(0)} = 0.1$, $\beta^{(0)} = 10^{-4}$, and $k = 0$.

while not converge **do**
  Updating $\mathcal{Z}^{(k)}$ via (13);
  Updating $\mathcal{H}^{(k)}$ via (16);
  Updating all $\mathbf{M}_n^{(k)}$ via (18);
  Updating all $\mathbf{V}_n^{(k)}$ via (20);
  Updating the multipliers $\mathbf{T}_n^{(k)}$ and $\mathcal{J}_1^{(k)}$ via (21);
  $\alpha^{(k)} = 1.05\alpha^{(k-1)}$, $\beta^{(k)} = 1.05\beta^{(k-1)}$ and $k = k + 1$;
end while

Output: HR-HSI $\mathcal{Z}$

TABLE I
AVERAGE QUANTITATIVE INDICES ON THE PAVIA UNIVERSITY DATASET

Methods	PSNR	SAM	ERGAS	UIQI	Time (s)
Ideal	∞	0	0	1	0
HySure [7]	42.0342	2.7289	2.2652	0.9720	288.74
STEREO [18]	43.4424	2.4526	1.7255	0.9733	32.00
LTMR [13]	44.0147	2.4085	1.4184	0.9774	154.43
SSR-Net [12]	43.8783	2.5078	1.5048	0.9766	**0.4150**
Proposed	**44.4120**	**2.3342**	**1.3892**	**0.9852**	18.32

run under MATLAB R2021a with Intel Core i7-10510U CPU with 16 GB RAM. SSR-Net [12] is implemented by Pytorch 1.6.0 on Python 3.6. The computing equipment contains 16 GB CPU memory and a GPU of GeForce GTX 1080Ti.

A. Experimental Results on Synthetic Dataset

The synthetic experiment is conducted on the Pavia University dataset. The dataset was collected by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor over the urban area of the University of Pavia, Italy. The full image is of size 610 × 340 × 115 with a spatial resolution of 1.3 m. After removing the water-vapor-absorption and noisy bands, it contains 93 spectral bands. We choose 600 × 320 pixels for the experiment.

In the synthetic experiment, original HSIs are treated as reference images for comparison convenience. The LR-HSIs are simulated by a 7 × 7 Gaussian blur kernel and then downsampled by a factor of 4, and the HR-MSIs are obtained through an IKONOS-like reflectance spectral response filter.

Experimental results are shown in Fig. 2, where Fig. 2(g) shows the reference image. Fig. 2(b)–(f) shows the pseudo color images of the HySure [7], STEREO [18], LTMR [13], SSR-Net [12], and the proposed method, respectively. To better visualize the spatial-spectral differences among compared methods, we also show the zoomed-in results of the same regions of interest together with the difference images computed from the reference image and different fusion results.

From the images shown in Fig. 2, we can clearly observe that all the methods can successfully improve the resolution of the LR-HSI. However, it still has some discrepancies when compared with the reference image. For instance, the HySure method [7] cause some spectral and spatial distortion as shown in Fig. 2(b). The results of STEREO [18] and SSR-Net [12] lose some spatial details, which can be seen from the difference maps. LTMR [13] can well improve the spatial quality, but from the amplified areas shown in Fig. 2(d), we can see that spatial details are missed to some degree. In contrast, the proposed method can better reduce the blurring effects while preserving image edges much sharper so as to perform much better results in preserving spectral information and providing high-quality spatial details.

Table I provides the quantitative results of different methods. The best result for each quantitative metric is in boldface. As listed in Table I, the proposed method gives the best results for the Pavia University dataset. Furthermore, the last column of Table I shows the running time of different methods on this data. It can be observed that the proposed method performs the best spectral and spatial qualities in a short time.

B. Experimental Results on Real Dataset

The dataset used in this experiment is HypSen. It consists of an LR-HSI with the spatial resolution of 30 m, and an HR-MSI with the resolution of 10 m. The LR-HSI was collected by Hyperion sensor with 89 bands, while the HR-MSI was taken

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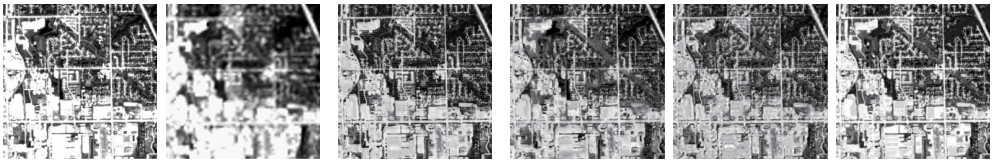

(a) (b) (c) (d) (e) (f)

Fig. 3. Experimental results of HypSen data (Band 1). (a) HR-MSI. (b) LR-HSI. (c) HySure [7]. (d) STEREO [18]. (e) LTMR [13]. (f) Proposed.

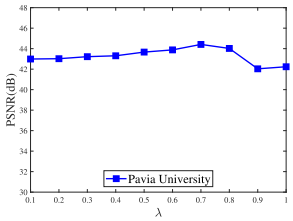

Fig. 4. PSNR curves as a function of λ

by the Sentinal-2A satellite with 4 spectral bands. In the experiment, we crop a subimage of 240×240 from MSI and a corresponding HSI subimage with a size of 80×80 for fusion.

Experimental results are shown in Fig. 3. It should be noted that training of SSR-Net [12] requires the reference images, and there is no reference for training, therefore, SSR-Net [12] is excluded in the experiment. We can observe that the proposed method provides more spatial details without the loss of spectral information.

C. Parameters Analysis

In the proposed method, λ is a regularization parameter, and we test our method for different values of λ. Fig. 4 plots the PSNR variations with respect to different λ on the Pavia University dataset. It can be observed from Fig. 4 that PSNR increases for the Pavia University dataset when λ varies from 0.1 to 0.7, and then it has an obvious drop when $\lambda = 0.8$. Therefore, we choose $\lambda = 0.7$.

IV. Conclusion

In this letter, we propose a variational tensor subspace decomposition-based fusion method. It explores the smoothness both in the spatial and spectral domains and learns the spatial-spectral differences and correlations by subspace learning. Experimental results on synthetic and real datasets show that the proposed method can obtain a better result with low computational complexity. However, the subspace learning is deduced from the Tucker decomposition, which is not suitable for large-scale tensors or high-order tensors. Therefore, further researches about the combination of tensor representation and subspace learning should be explored in the future.

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